



Natural Language Processing: Classification

HSE Faculty of Computer Science
Machine Learning and Data-Intensive Systems

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- **General view on classification with text**
- Pre-deep learning approaches
- Deep learning approaches
- Practical Tips

Extract features, apply model, compare the distribution to the target

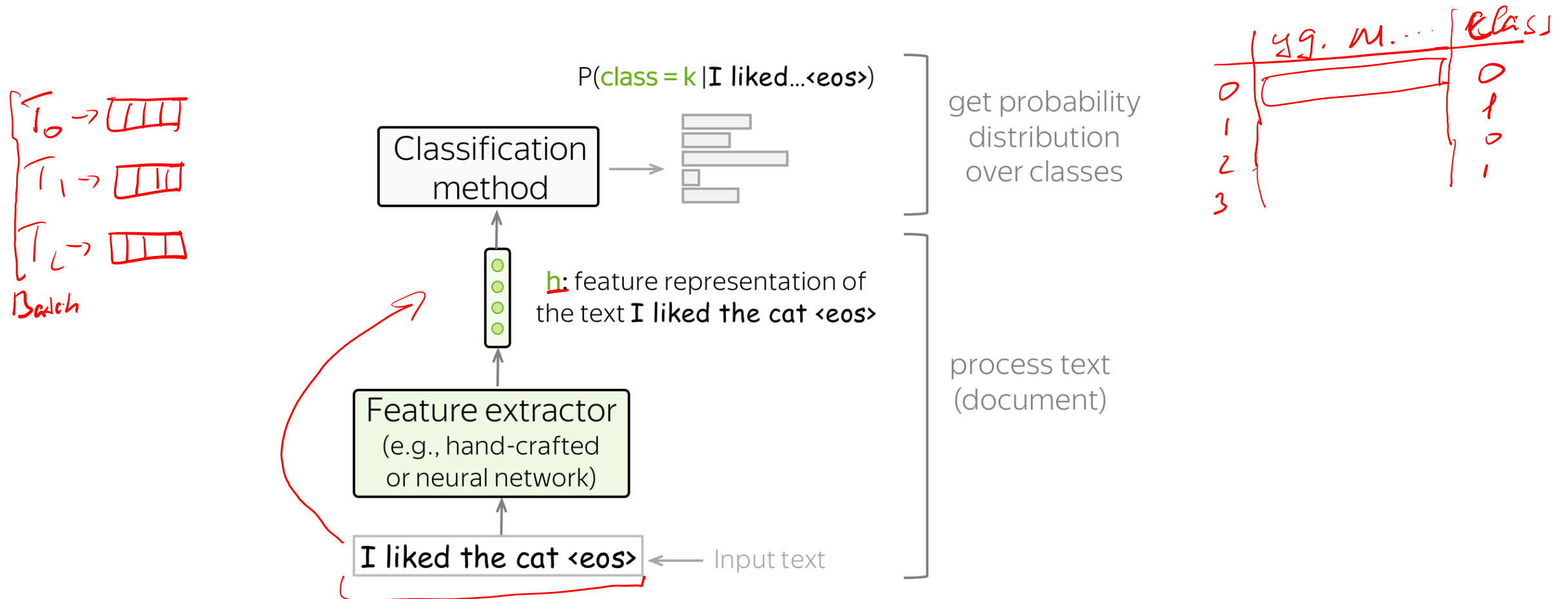
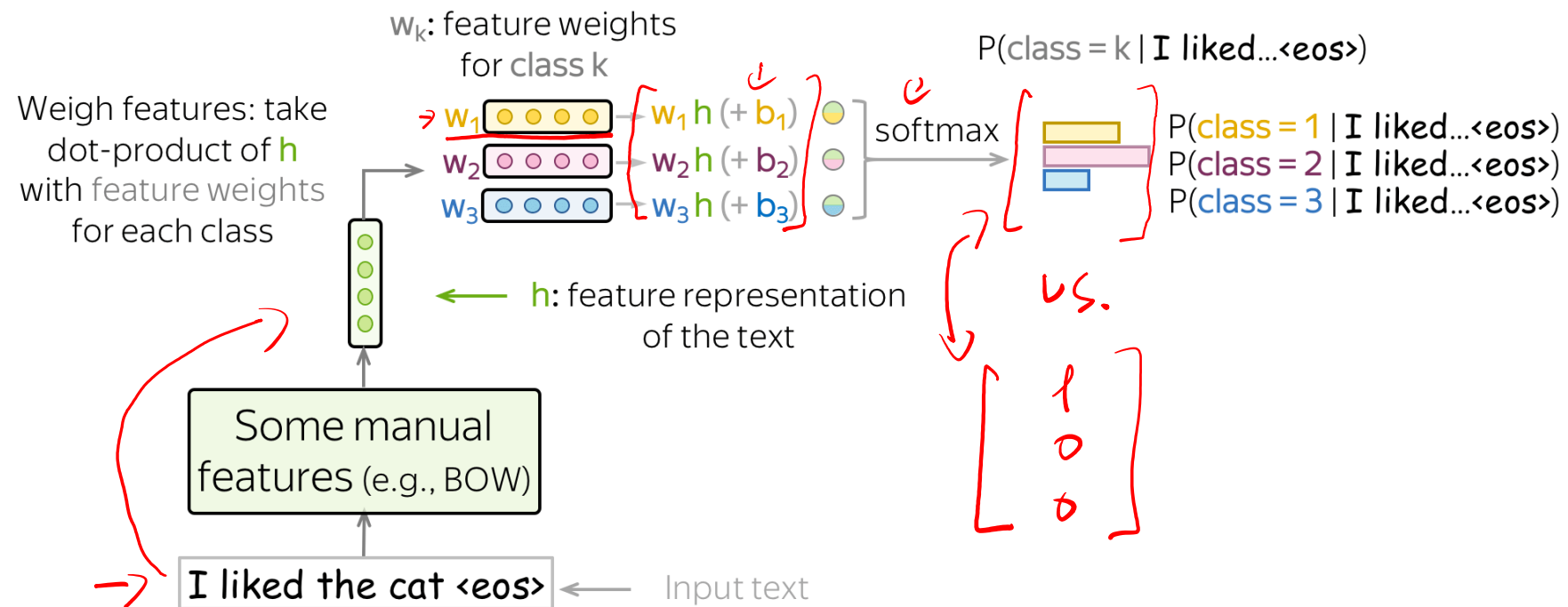




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- Deep learning approaches
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Train a logistic regression



Loss function: log-loss

Given training examples $x^1, \dots, x^N$ with labels $y^1, \dots, y^N$, $y^i \in \{1, \dots, K\}$, we pick those weights $w^{(k)}$, $k = 1..K$ which maximize the probability of the training data:

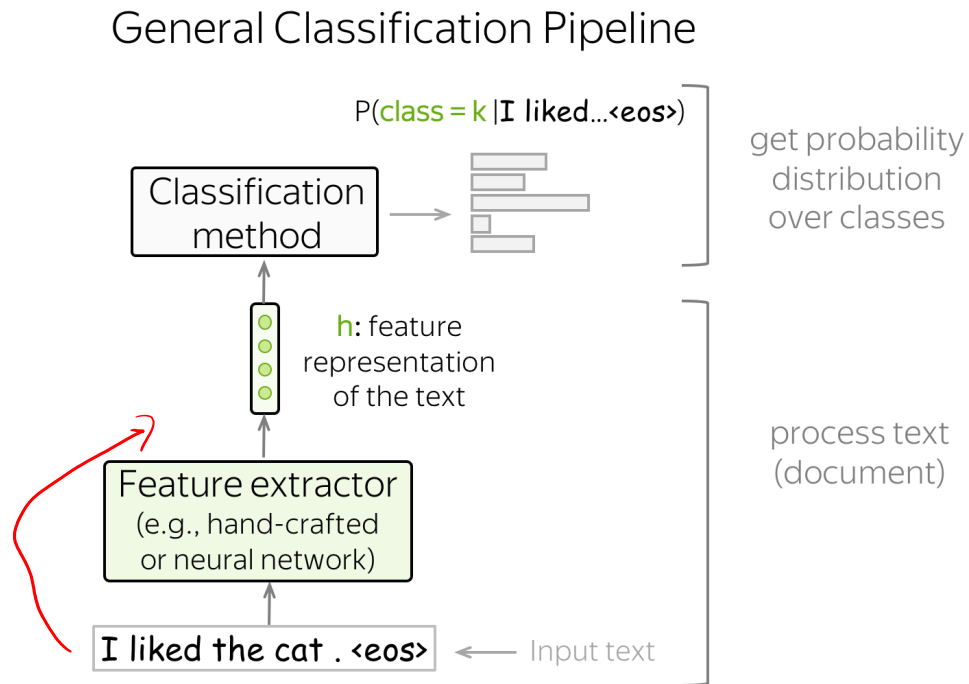
$$w^* = \arg \max_w \sum_{i=1}^N \log P(\underline{y} = \underline{y^i} | x^i).$$



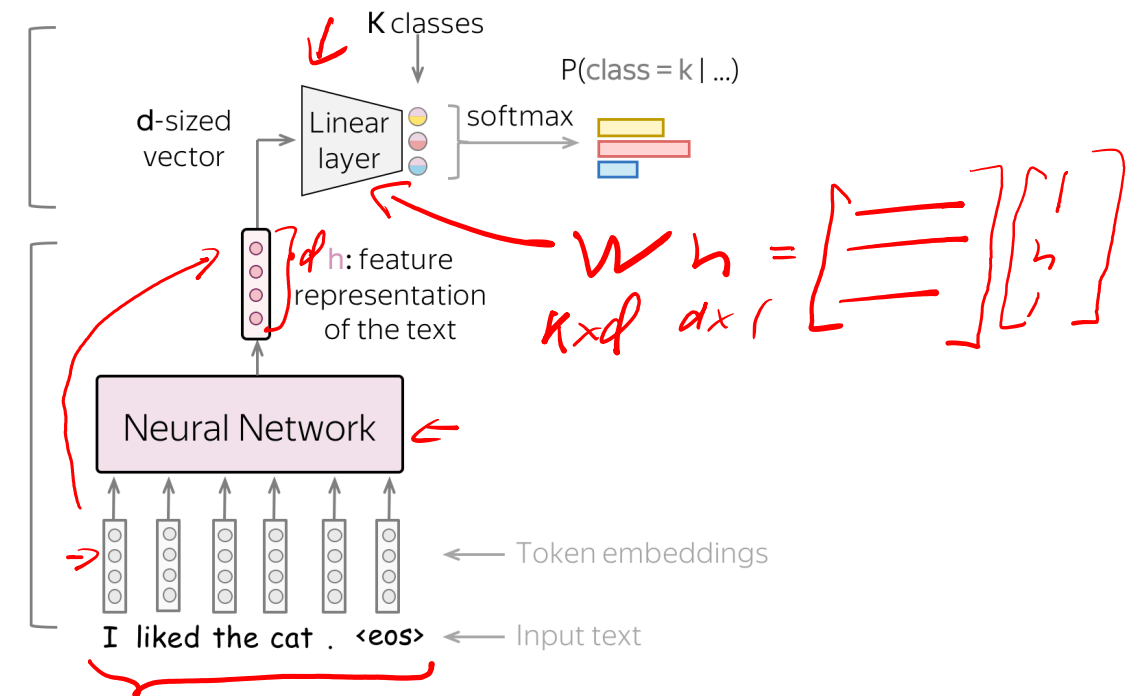
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Overview of DL-based approaches



Classification with Neural Networks



Overview of DL-based approaches

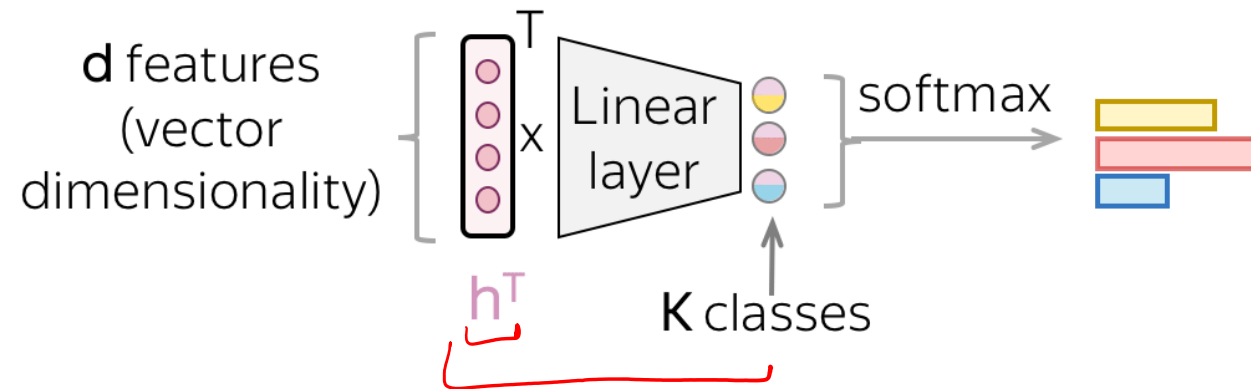
We have:

- h - vector of size d

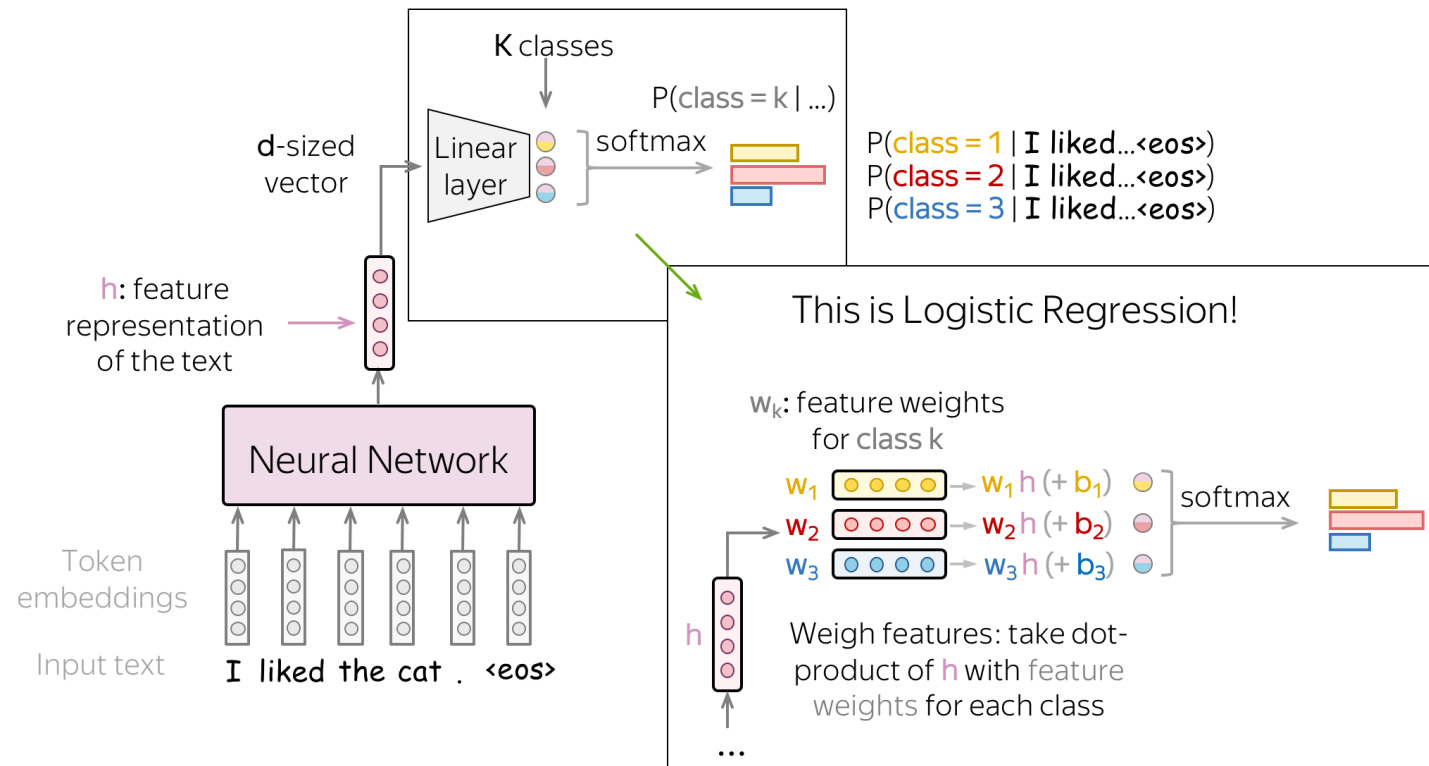
We need:

- vector of size K – probabilities for K classes

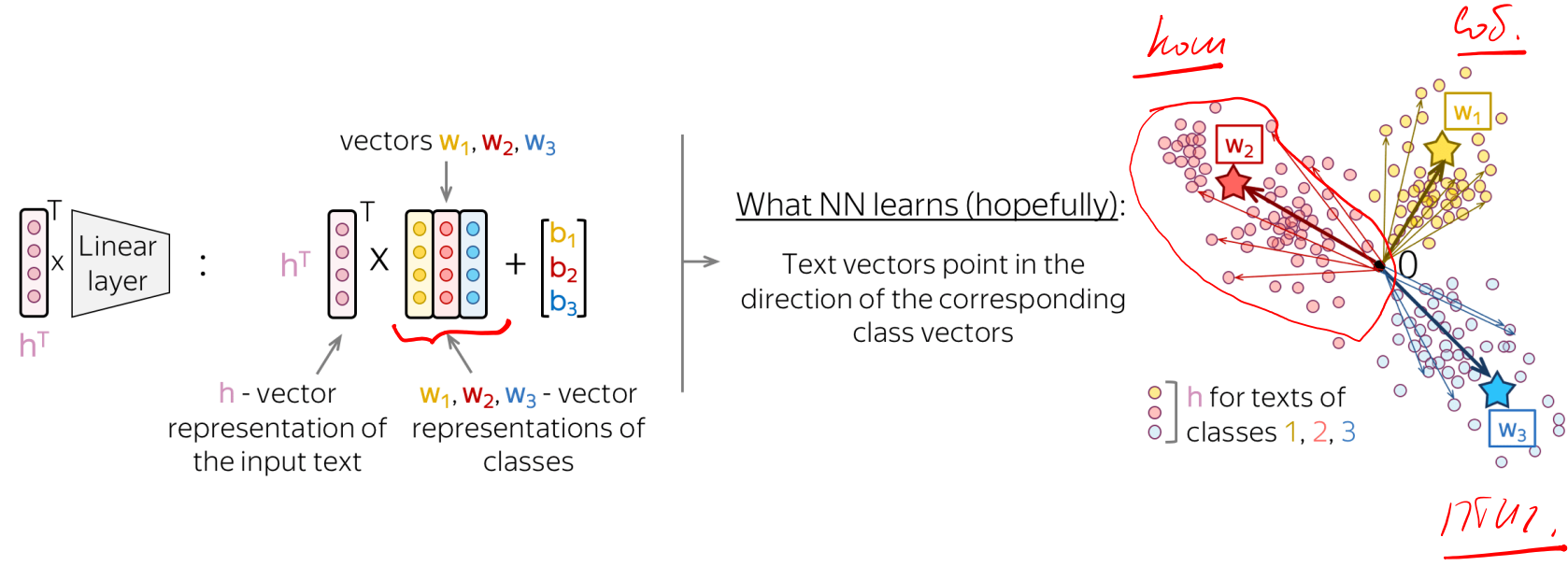
Transform linearly
from size d to size K



Overview of DL-based approaches



Overview of DL-based approaches



Overview of DL-based approaches

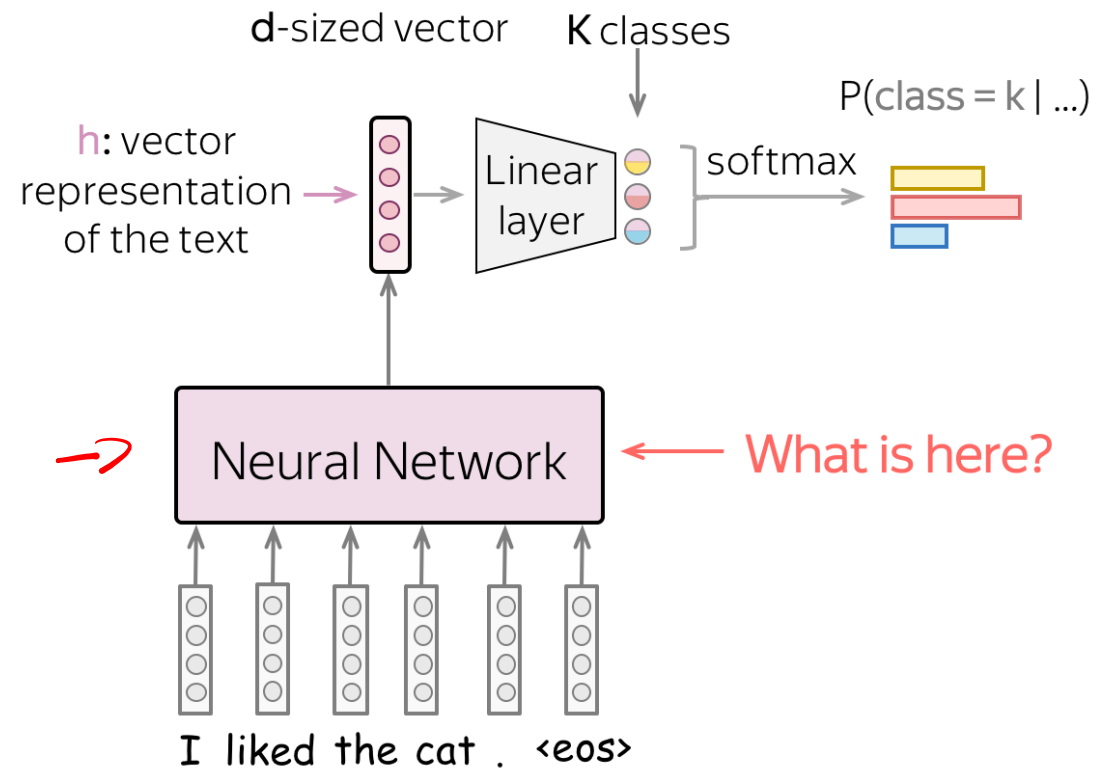
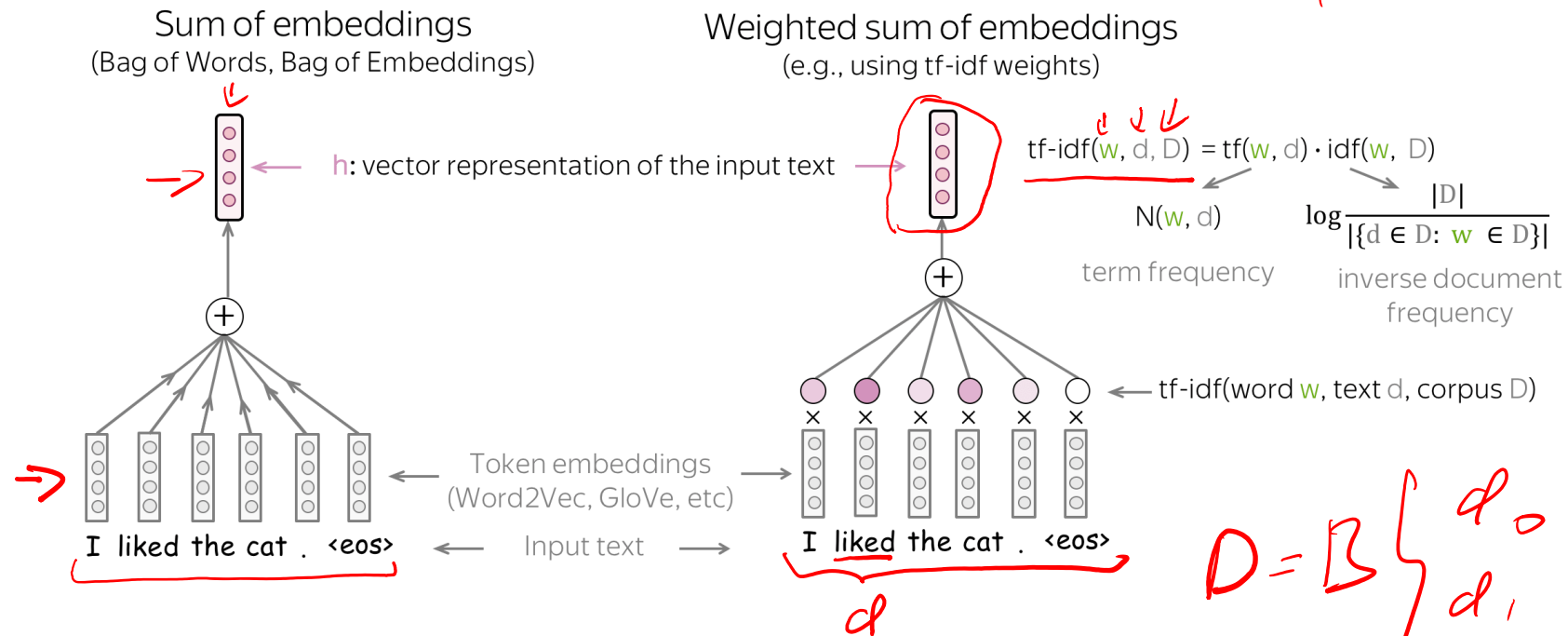




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Vanilla embeddings



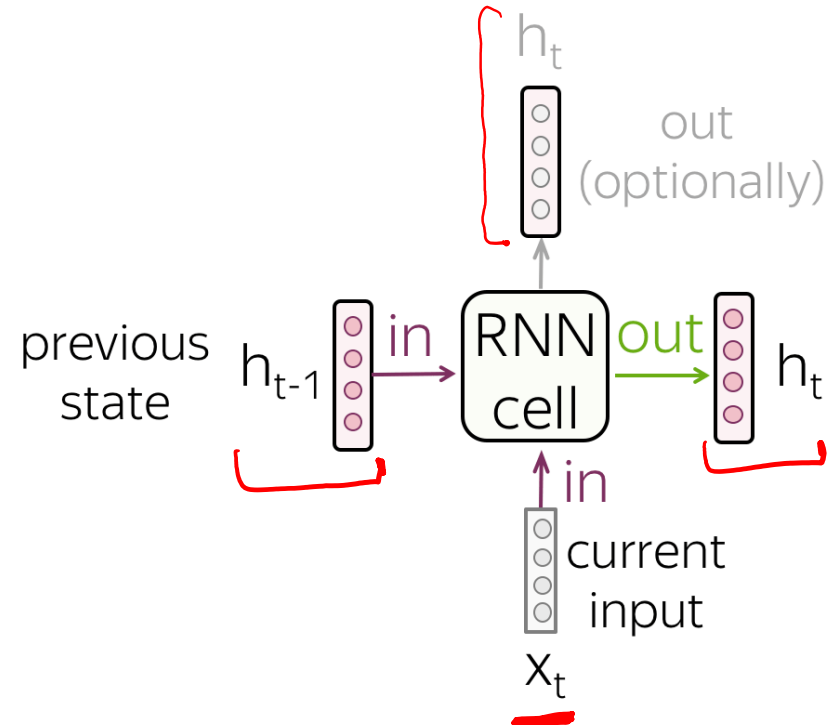
$$\begin{array}{c|ccc} & d_0 & d_1 & d_2 \\ \hline w_0 & \cdot & \cdot & \cdot \\ w_1 & \cdot & \cdot & \cdot \\ w_2 & \cdot & \cdot & \cdot \end{array}$$



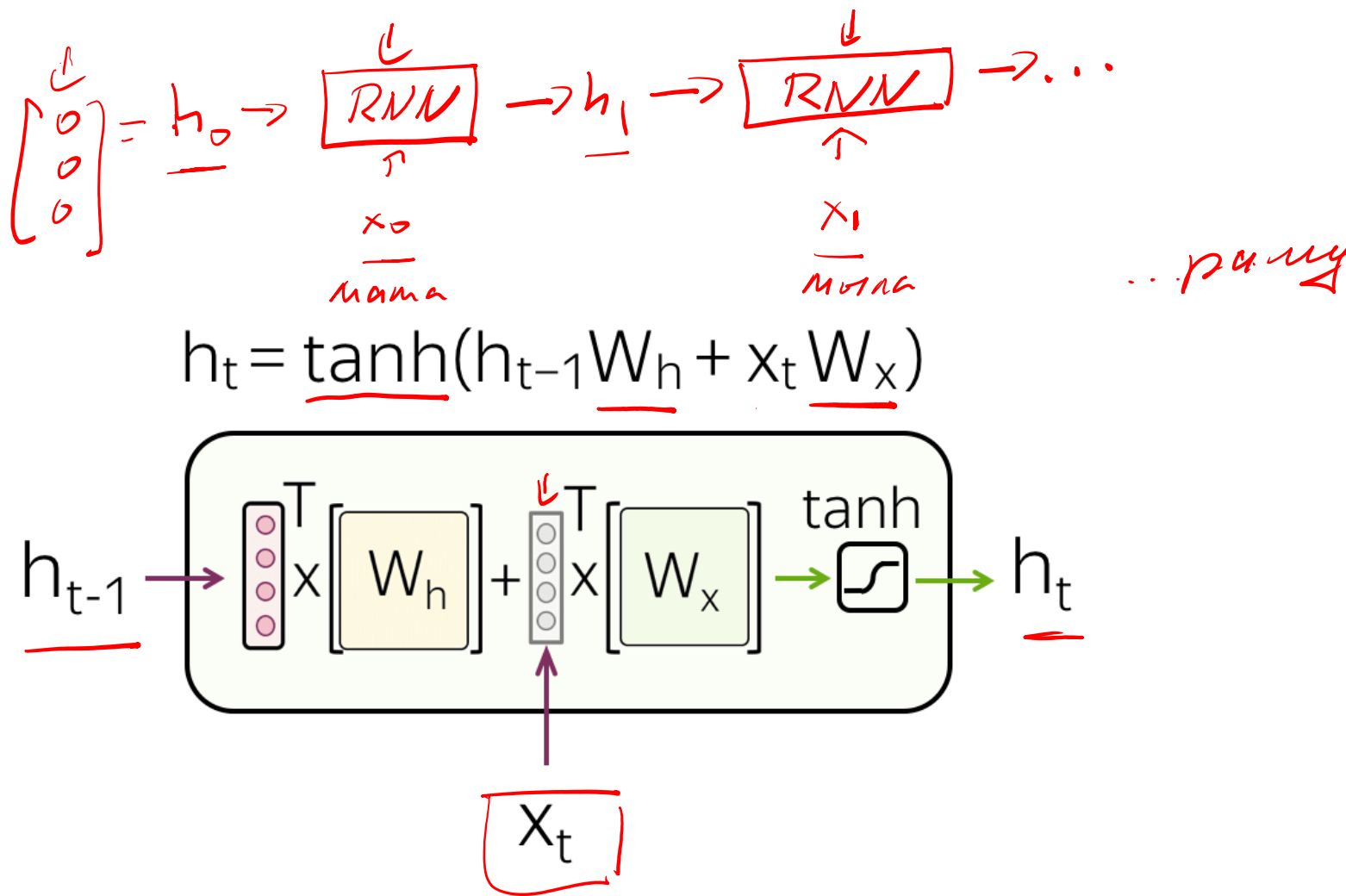
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- **Deep learning approaches: RNN**
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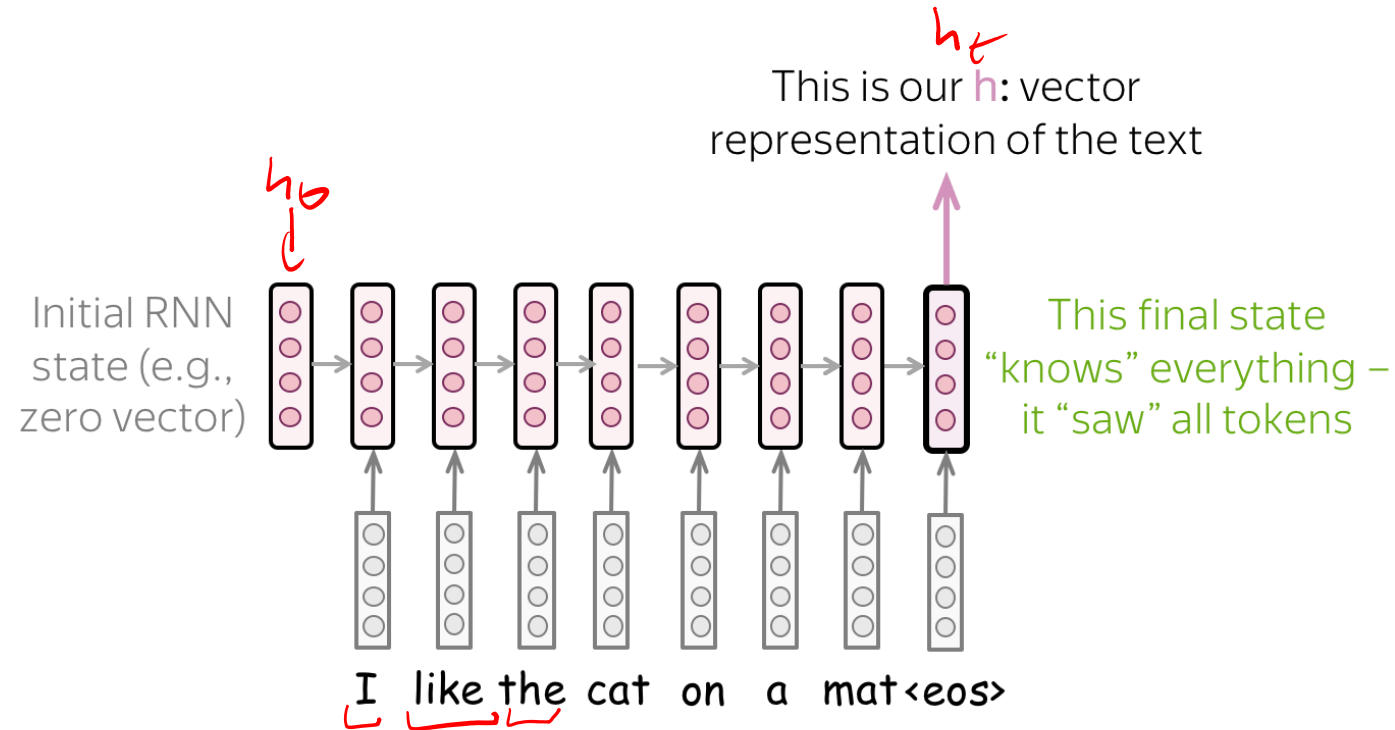
Vanilla embeddings



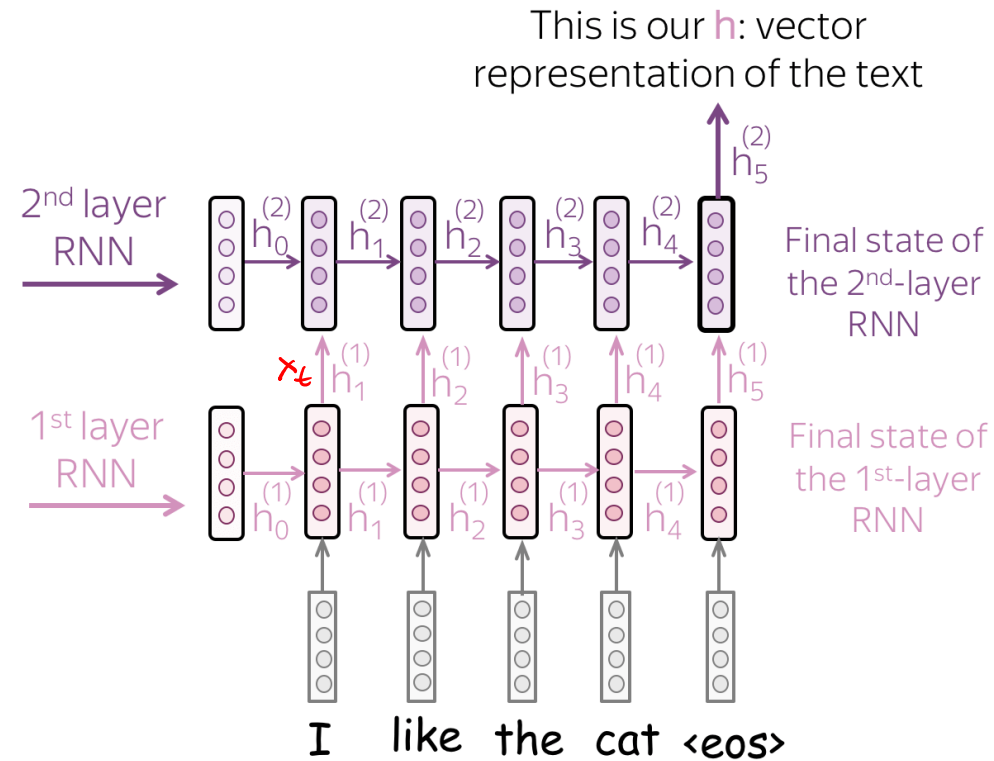
Vanilla RNN



Vanilla RNN



Stacked RNN



Bidirectional RNN

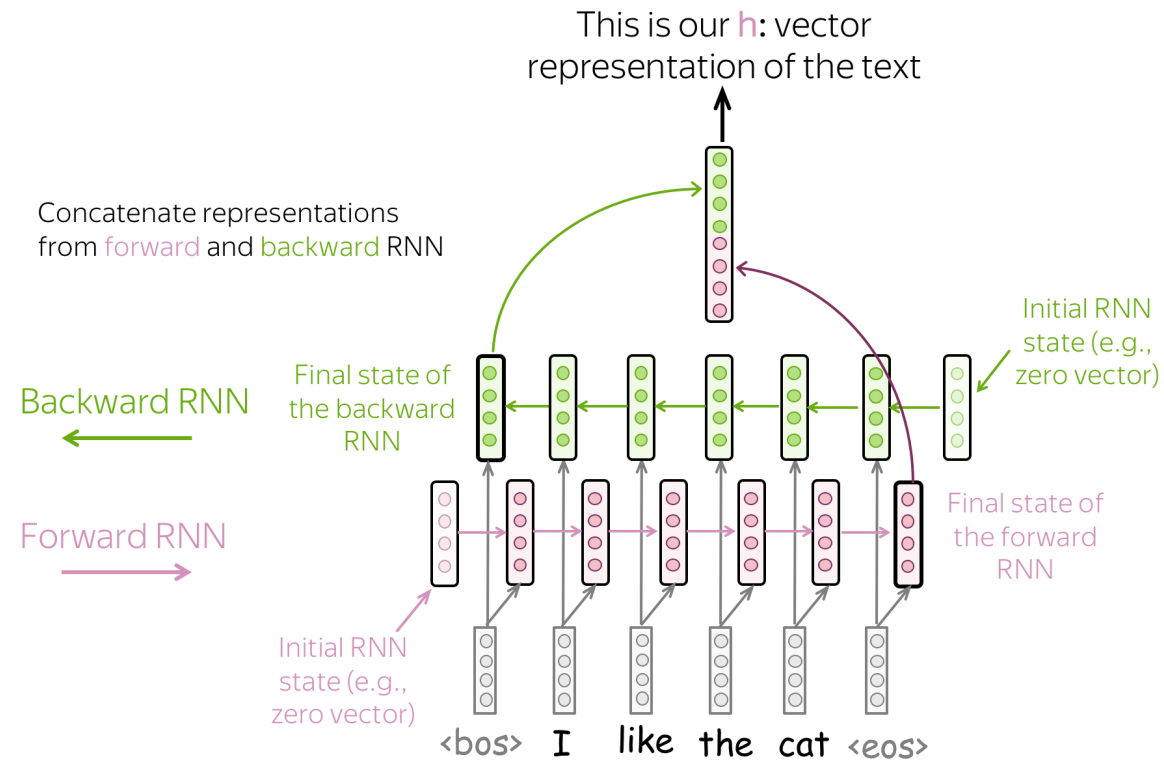
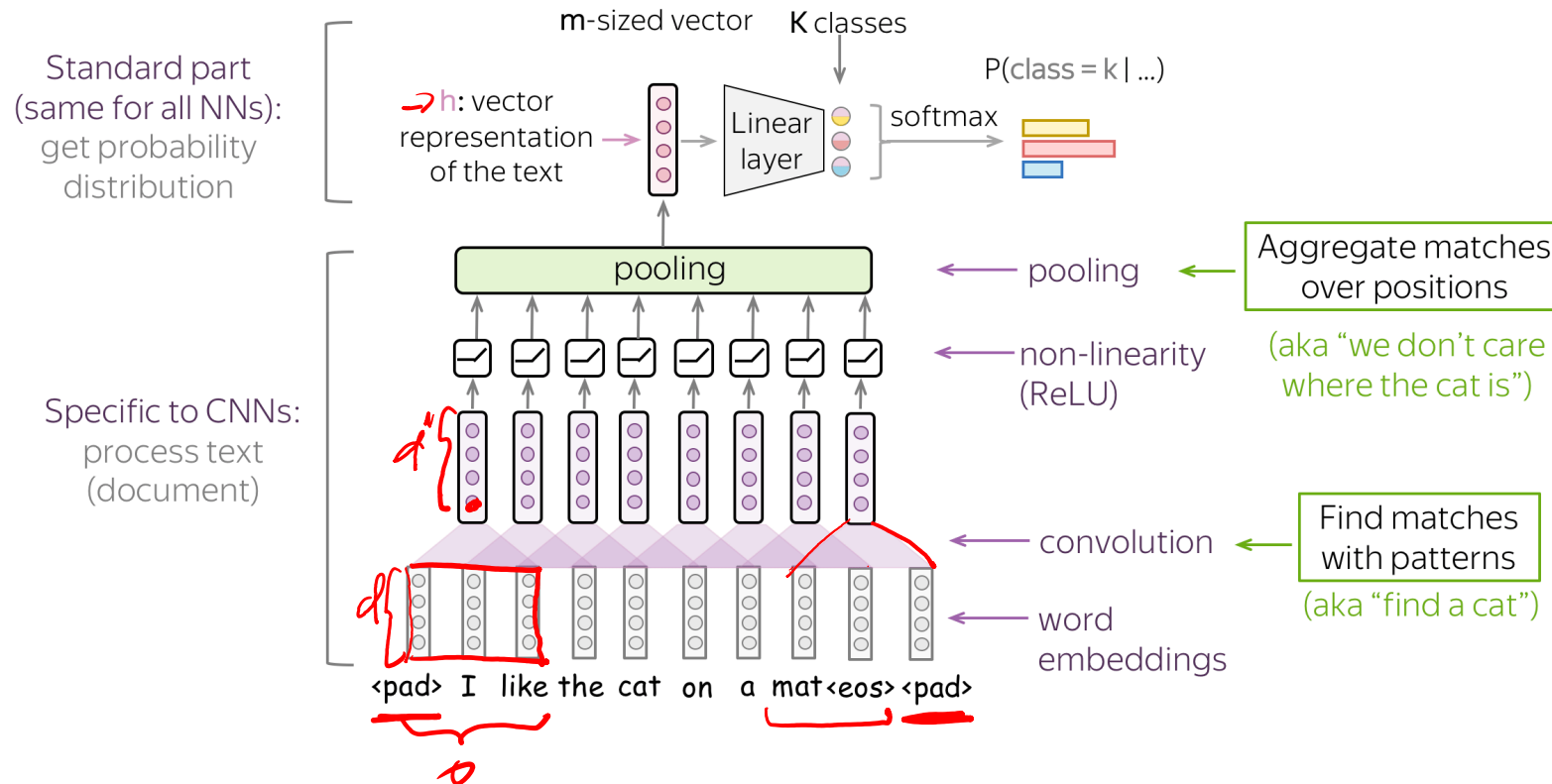




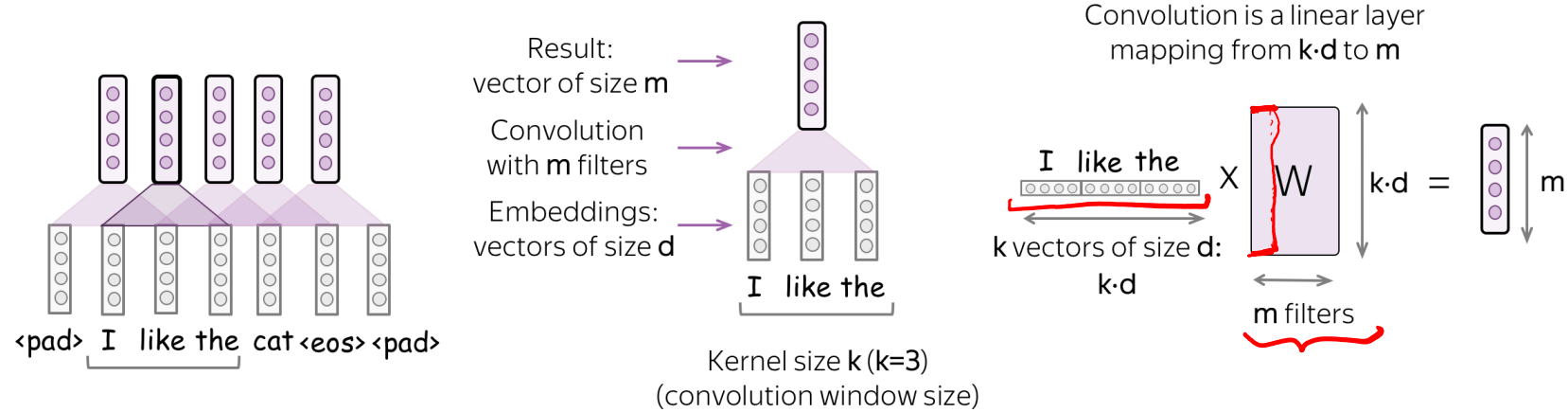
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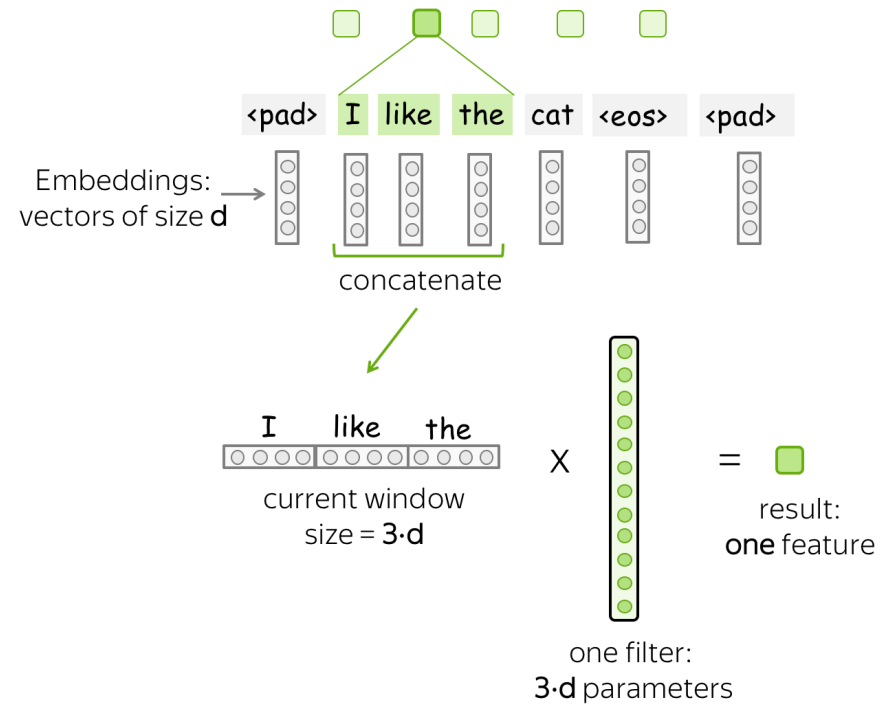
CNN overview



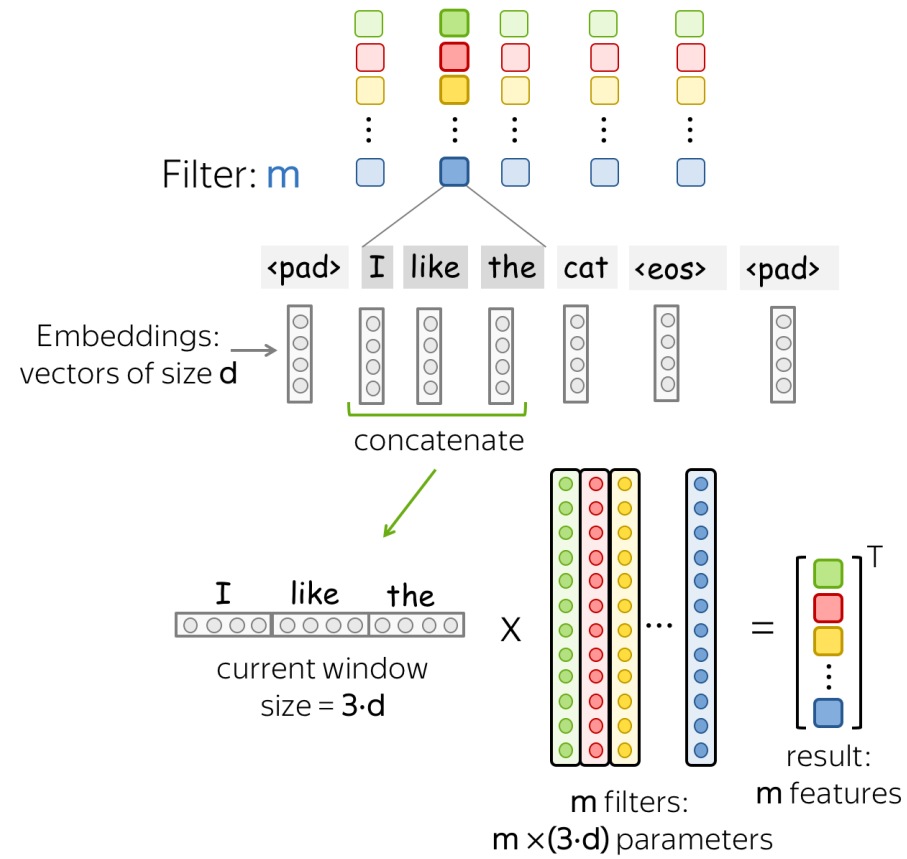
Convolution is a Linear Operation Applied to Each Window



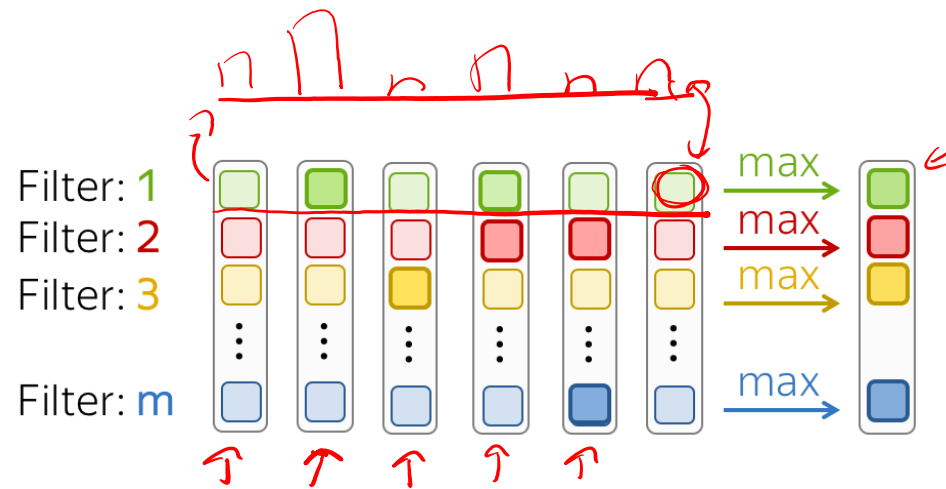
CNN intuition



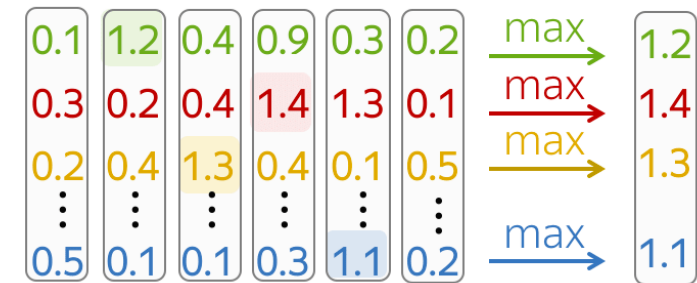
CNN intuition



Pooling



Max pooling:
maximum for each
dimension (feature)



Larger kernel size for a broader context

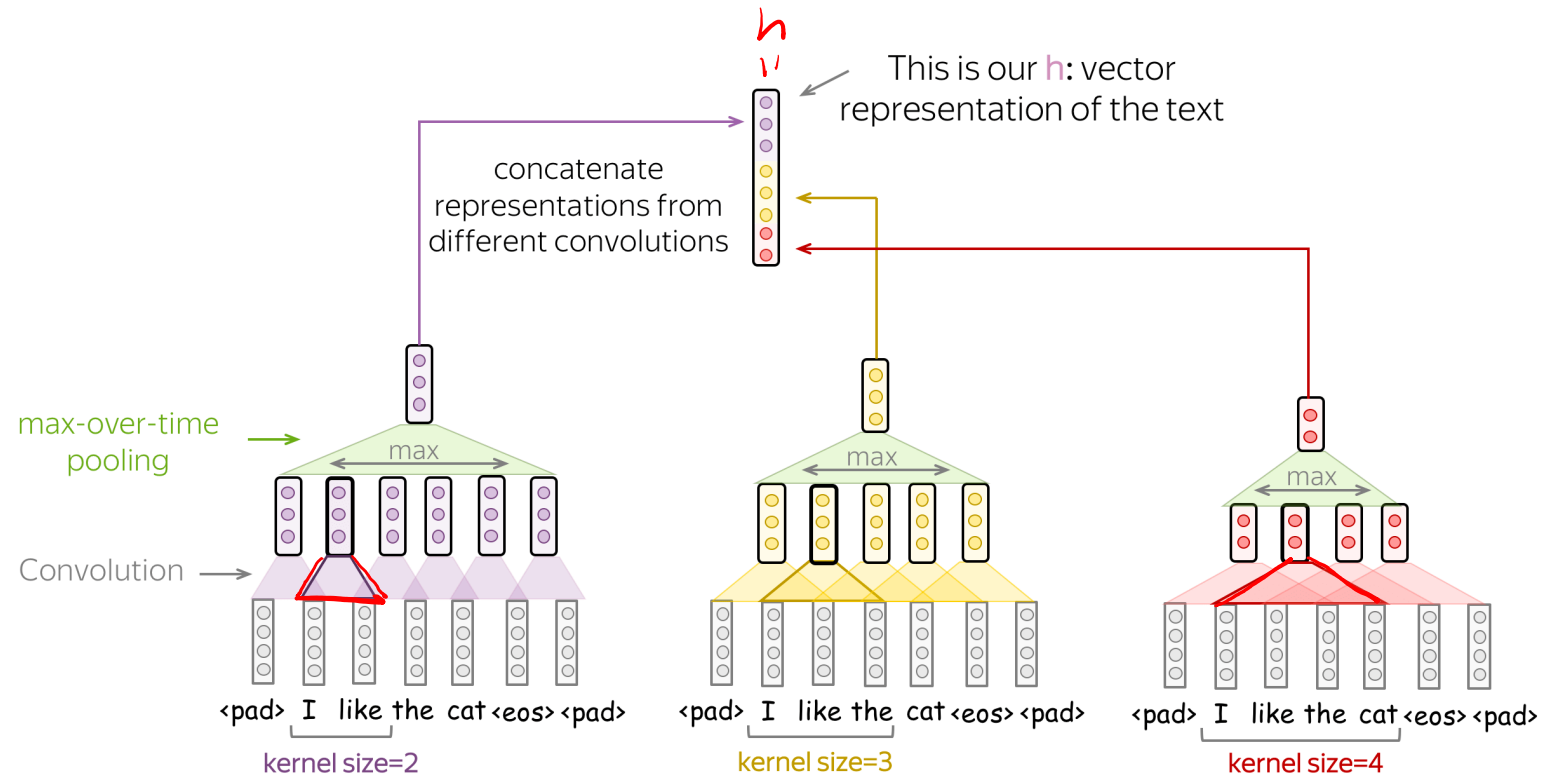
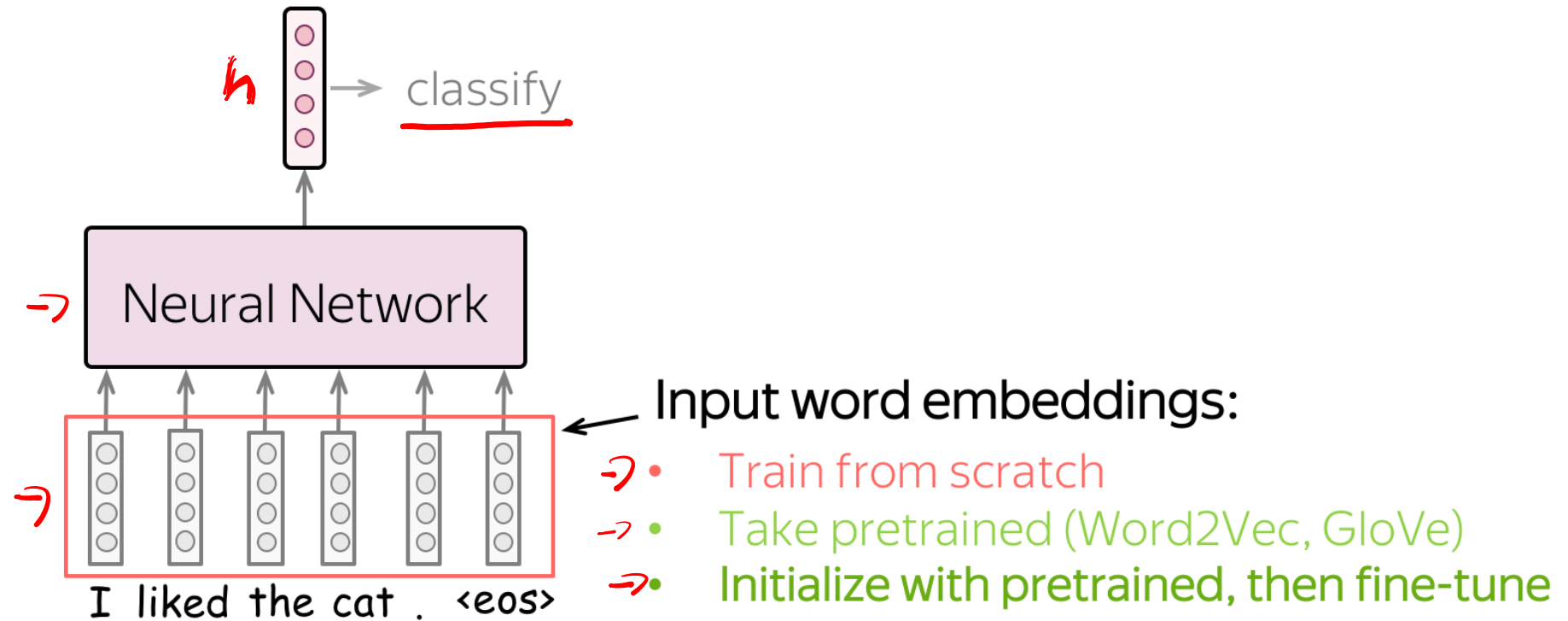




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Larger kernel size for a broader context



Data augmentation

